

## BIOMECHANICS (EPHE 341)

# Dynamic Analysis

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Last week every equation had zero on the right-hand side. This week the zeros become  $ma$  and  $I\alpha$ , and nothing else changes.

## Where this fits

Newton's first law gave us two cases, and last week we took the first one.

- $\Sigma F = 0$  and  $\Sigma M = 0$  — nothing is accelerating. **Static analysis.**
- $\Sigma F \neq 0$  or  $\Sigma M \neq 0$  — the state of motion is changing. **Dynamic analysis.**

Almost everything a person does is the second case. Walking is the second case. So is standing up, throwing, landing, and pedalling.

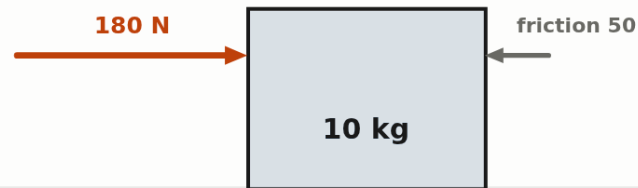
A body moving at **constant velocity** is still the first case. No acceleration means no net force, however fast it is going. His slide 2 calls this *dynamic equilibrium*, and it is genuinely a statics problem wearing a coat.

The phrase comes from a trick: put the quantity  $ma$  on the diagram as though it were a force, pointing backwards, and the sums are zero again. That is d'Alembert's idea, and it is why the method does not have to change.

## The right-hand side

$$\Sigma F_x = m \cdot a_x$$

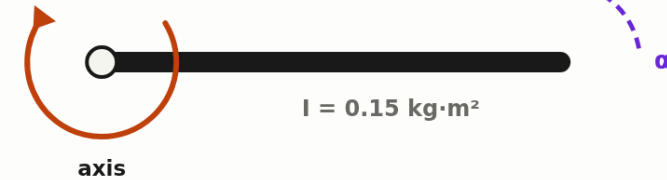
$$m \cdot a = 130 \text{ N}$$

$$180 - 50 = 10 \cdot a$$
$$a = 13.00 \text{ m/s}^2$$

$$\Sigma M = I \cdot \alpha$$

$$M = 12 \text{ N}\cdot\text{m}$$



$$12 = 0.15 \cdot \alpha$$
$$\alpha = 80.0 \text{ rad/s}^2$$

The free body diagram does not change · the force table does not change · the sign conventions do not change · the only thing that changes is that the right-hand side is no longer zero

The push

180 N

The twist

12 N·m

Last week the right-hand side was zero and you solved for the forces that made it so. This week it is  $m \cdot a$  and  $I \cdot \alpha$ , and you solve for the forces that produce the motion you measured. Slide the push down to 50 N, where it exactly cancels friction, and the dynamic problem turns back into the static one without a single line of the method changing. The dashed arrow is the quantity  $m \cdot a$  drawn as though it were a force: add it and the sums are zero again, which is why his slide 2 calls this **dynamic equilibrium**.

*The two equations, in general form*

$$\Sigma F = m \cdot a$$

Newton's second law. The net force on a body equals its mass times the acceleration of its **centre of mass** — not of any other point on it.

$$\Sigma M = I \cdot \alpha$$

The rotational twin. The net moment equals the moment of inertia times the angular acceleration, both taken about the **same axis**.

In two dimensions that is three scalar equations, and three is exactly what a free body diagram gives you:

$$\Sigma F_x = m \cdot a_x$$

$$\Sigma F_y = m \cdot a_y$$

$$\Sigma M = I \cdot \alpha$$

Three equations means **three unknowns**, and not one more. Everything about how inverse dynamics is organised follows from that one sentence.

## *Leonhard Euler, 1707–1783*

Newton gave us  $F = ma$  for a particle. It was **Euler** who wrote the rotational law for a rigid body and showed that the two together are enough to describe any motion of anything solid.

Hence **Newton–Euler** equations, which is what the method in this lecture is properly called, and what you will see it called in every paper that uses it.

Euler also lost the sight in one eye at 28 and in the other at 59, and produced roughly half of his work after going completely blind, dictating it to his sons. He is still the most prolific mathematician who has ever lived.

## *Why two dimensions*

Everything in this lecture is **planar**: motion in the sagittal plane, forces in the sagittal plane, one moment axis perpendicular to it.

In three dimensions there are six equations rather than three, and the moment equation acquires terms that have no two-dimensional equivalent. The *method* is identical. The bookkeeping is not.

This is not only a teaching simplification. A large fraction of published gait analysis reports sagittal joint moments computed exactly this way, because the sagittal plane is where most of walking happens.

At the end of the lecture we will run the real thing on a real stride and find out how much the simplification costs.

# **Two directions**

Forces in, motion out — or motion in, forces out. One of them you can do to a person.

## *Forward and inverse*

**Forward dynamics.** You know the forces and the moments. You solve for the accelerations, and from them the motion.

This is what a simulation does, and what a robot's controller does. It is also what every physics problem you have ever been set does.

forces  $\rightarrow a, \alpha \rightarrow$  motion

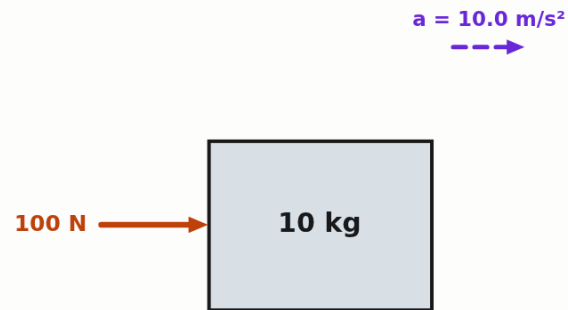
**Inverse dynamics.** You know the motion, because you filmed it, and you know the external forces, because you stood the person on a plate. You solve for the forces and moments **inside**.

This is the one you can do to a human being, and it is the rest of this lecture.

motion + external forces  $\rightarrow$  internal forces

The asymmetry is the whole point. You cannot put a load cell inside somebody. You can put a camera in front of them and a force plate under them — so the only direction available is the inverse one.

## Forward dynamics, four examples



A force of 100 N on a 10 kg object. What is the acceleration?

$$\Sigma F = m \cdot a$$

the forces all act through the centre of gravity, so nothing turns

$$\Sigma F = 100 = 100 \text{ N}$$

$$a = 100 / 10 = 10.00 \text{ m/s}^2$$

his answer too

Forward dynamics: the forces are given and the motion comes out. This is the direction a simulation runs, and the direction you can almost never use on a person, because the forces inside are the part you cannot measure.

#1

#2

#3

#4

Four problems, one equation each, and between them they are the whole of forward dynamics: **known forces in, motion out**. Two linear, two angular. Nothing here needs a free body diagram with unknowns in it, which is exactly what makes the inverse problem — the rest of this lecture — harder.

## *Where forward dynamics does earn its keep*

Build a model of a body — segments, joints, muscles with force–length and force–velocity properties — choose a pattern of muscle activation, and let the equations run. The model walks, or jumps, or falls over.

Then change something you could never change in a person: delete a muscle, weaken a tendon, alter a moment arm, add a prosthesis. Run it again.

Forward dynamics answers questions of the form *what would happen if*. Inverse dynamics answers *what did happen*. You need both, and they are not substitutes.

The cost is that a forward model is only as good as the assumptions inside it, and there are hundreds of them. An inverse-dynamics answer rests on far fewer — which is the subject of the last third of this lecture.

# **The method**

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Break the limb into segments, start at the one touching the ground, and carry the answer upward.

## *What you can actually measure*

We can record the **movement**: a motion capture system, video, accelerometers, gyroscopes.

We can record the **external forces**: a force plate gives the ground reaction force and where it acts.

We cannot record anything **inside**. Not one tendon, not one joint contact force, not one ligament.

So: recorded trajectories, recorded ground forces, Newton–Euler, and out comes the **net joint moment and the net joint force** at every joint, at every instant.

The word **net** is doing a great deal of work in that sentence, and we will come back to it before the end.

## Why you have to start at the foot



### starting at the most distal segment

1. the foot 3 equations · 3 unknowns

the force at the far end is the one the plate measured

2. the shank 3 equations · 3 unknowns

the far end came out of step 1, so only the knee is left

3. the thigh 3 equations · 3 unknowns

the far end came out of step 2, so only the hip is left

Three segments, three solutions, each one handing the next the forces and the moment it needs. This is his slides 17 and 18.

Start at the foot

Start at the hip

Step 1

Step 2

Step 3

A free body in two dimensions gives you **three** equations:  $\sum F_x$ ,  $\sum F_y$ ,  $\sum M$ . An unknown joint costs **three** unknowns: two force components and a moment. So a segment can be solved only when one of its two ends is already known — and at the start of the problem exactly one end of one segment is: the foot's contact with the ground, where a force plate has measured it.

## *The foot, cut free*

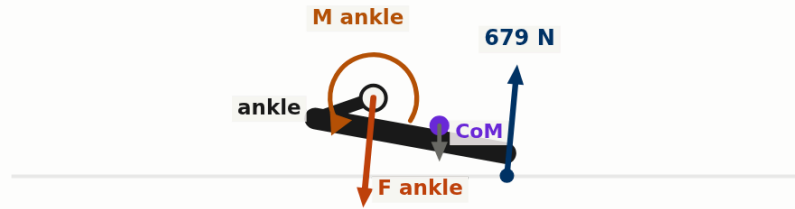
Draw the foot on its own and put on it **everything** that touches it:

- the **ground reaction force**, acting at the centre of pressure — measured;
- its own **weight**, acting at its centre of mass — from the anthropometric table;
- the **force the shank applies** at the ankle, two components — unknown;
- and the **moment the shank applies** at the ankle — unknown.

Three unknowns, three equations. The two force sums come **first**, because the ankle force appears in the moment equation and you need its value before you can use it. That is the only ordering rule in the whole method.

His slide 19 makes the simplification explicit: every muscle, ligament and contact force crossing the ankle is replaced by **one net force and one net moment**. That replacement is exact for the mechanics and lossy for the anatomy.

## The foot, on a real stride



44 % of the gait cycle

$$\Sigma F_x = m \cdot a_x$$

$$F(\text{ankle})_x + 63 = 0.80(0.79)$$

$$F(\text{ankle})_x = -62.1 \text{ N}$$

$$\Sigma F_y = m \cdot a_y$$

$$F(\text{ankle})_y + 676 - 7.8 = 0.80(1.33)$$

$$F(\text{ankle})_y = -667.2 \text{ N}$$

$$\Sigma M = I \cdot \alpha$$

$$M(\text{ankle}) + M(F \text{ ankle}) + M(\text{GRF}) = 0.00381(-25.6)$$

$$M(\text{ankle}) = -94.48 \text{ N} \cdot \text{m}$$

that is 1.718 N·m per kilogram, counting a plantarflexor moment as positive

Winter's table gives the foot 0.80 kg and 0.00381 kg·m<sup>2</sup> · the markers give its acceleration · the plate gives the force and where it acts · everything else is the three equations

Where in the stride



44 %

The diagram

The forces

**The moment**

▶ Walk

His slides 20 to 24, with a real foot in them. The two force sums come first because the ankle's force has to be known before it can appear in the moment equation — the only ordering rule in the method. Walk past 60 % and the plate loses the foot: with no ground force the ankle moment collapses to almost nothing, which is why his swing example comes out at fractions of a newton metre while stance runs to a hundred.

## *The moment arms*

Take every moment about the segment's own **centre of mass**. Then  $I\alpha$  is the plain  $I\alpha$  from the table, with no parallel-axis correction.

His slides write each force's moment as two terms:

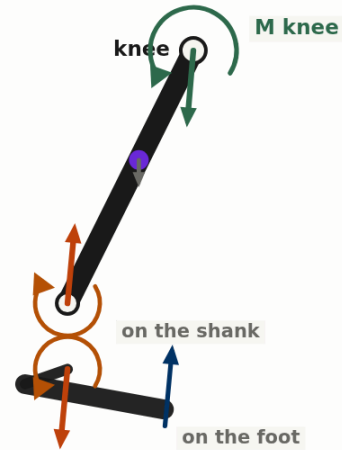
$$F_x \cdot d(\sin\theta) + F_y \cdot d(\cos\theta)$$

which is the same thing as the cross product  $r \times F$ , written out. The horizontal force acts through a **vertical** arm, and the vertical force through a **horizontal** one.

Pairing  $F_x$  with the horizontal distance is the single commonest error in this topic. The arm a force turns through is **perpendicular** to it.

His slides give the arms as positive lengths and assign the signs by looking at the picture. That works, and it is also where marks go missing. If you write the positions as signed vectors first and take  $r \times F$ , the signs look after themselves — including the reversal at the ankle that Newton's third law demands.

## Up the chain



44 % of the gait cycle

what the foot produced

$$\mathbf{F} = (-62.1, -667.2) \text{ N}, \quad \mathbf{M} = -94.48 \text{ N}\cdot\text{m}$$

reversed onto the shank

$$\mathbf{F} = (62.1, 667.2) \text{ N}, \quad \mathbf{M} = 94.48 \text{ N}\cdot\text{m}$$

$\Sigma F_x, \Sigma F_y$  at the knee

$$\mathbf{F}(\text{knee}) = (-54.1, -638.2) \text{ N}$$

$\Sigma M = I \cdot \alpha$  at the knee

$$\mathbf{M}(\text{knee}) = -2.40 \text{ N}\cdot\text{m} = -0.044 \text{ N}\cdot\text{m/kg}$$

Both halves of the handover are in: the force reversed and the moment reversed. At this instant dropping the moment alone would move the knee answer by 94.48 N·m.

Where in the stride



44 %

Hand the moment up

Forget the moment

► Walk

Newton's third law applies to moments as well as to forces. The foot's free body produced **two** things at the ankle, and **both** come back reversed on the shank. His slide 27 carries them as the two terms –  $\mathbf{M}(\text{ankle})$  and –  $\mathbf{F}(\text{ankle})$ , and the minus signs are not decoration.

## *The method, in six lines*

- Record the **kinematics** and any **external forces**.
- Get each segment's **mass, centre of mass and moment of inertia** from an anthropometric table.
- Differentiate the kinematics twice for  $a$  and  $\alpha$ .
- Start at the **most distal** segment and write its three equations.
- **Reverse** the force and the moment you found, and apply them to the next segment up.
- Repeat to the top of the chain.

That is his slide 28, and it is the examinable part. Everything after this is one worked example and then a long, honest look at how much the answer can be trusted.

# The worked example

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The ankle and the knee during the swing phase of walking, line by line.

# A Worked Example

## *What you are given*

|                                | Foot    | Leg    |
|--------------------------------|---------|--------|
| $m$ (kg)                       | 1.16    | 3.72   |
| $I$ (kg·m <sup>2</sup> )       | 0.00995 | 0.0642 |
| $a_x$ (m/s <sup>2</sup> )      | 9.07    | -0.03  |
| $a_y$ (m/s <sup>2</sup> )      | -6.62   | -4.21  |
| $\alpha$ (rad/s <sup>2</sup> ) | 21.63   | 36.9   |

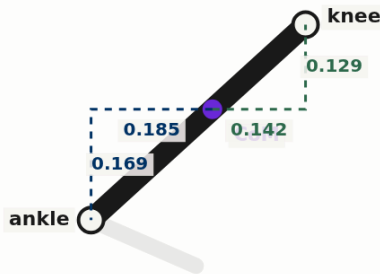
Two masses, two inertias, four accelerations, two angular accelerations, and the distances from each joint to each segment's centre of mass. Nothing else.

Worth checking where those masses came from. Winter's table gives the foot 1.45 % of body mass and the leg 4.65 %.  $1.16 / 0.0145 = \mathbf{80.0\ kg}$ .  $3.72 / 0.0465 = \mathbf{80.0\ kg}$ . The same person, to three figures.

The leg's inertia checks out too: it implies a radius of gyration of 0.297 of the segment length against Winter's 0.302, and his joint distances put the leg's centre of mass at 0.434 of its length against Winter's 0.433.

His worked example, every line

the swing phase · slides 30-41



4. the shank,  $\Sigma M = I \cdot \alpha$

the knee force about the CoM: 2.141 N·m

the ankle force about the CoM: -1.094 N·m

and the ankle moment, reversed: -0.39 N·m

$$M(\text{knee}) = I \cdot \alpha - (2.14) - (-1.09) - (-0.39) = 1.72 \text{ N}\cdot\text{m}$$

|               |               |
|---------------|---------------|
| M(F knee/CG)  | his slide too |
| M(F ankle/CG) | his slide too |
| M(knee)       | his slide too |

Four steps, two segments, and the whole of his worked example. Every number on his slides 30 to 41 reproduces exactly.

His swing example

His stance Example 2

Foot: forces

Foot: moment

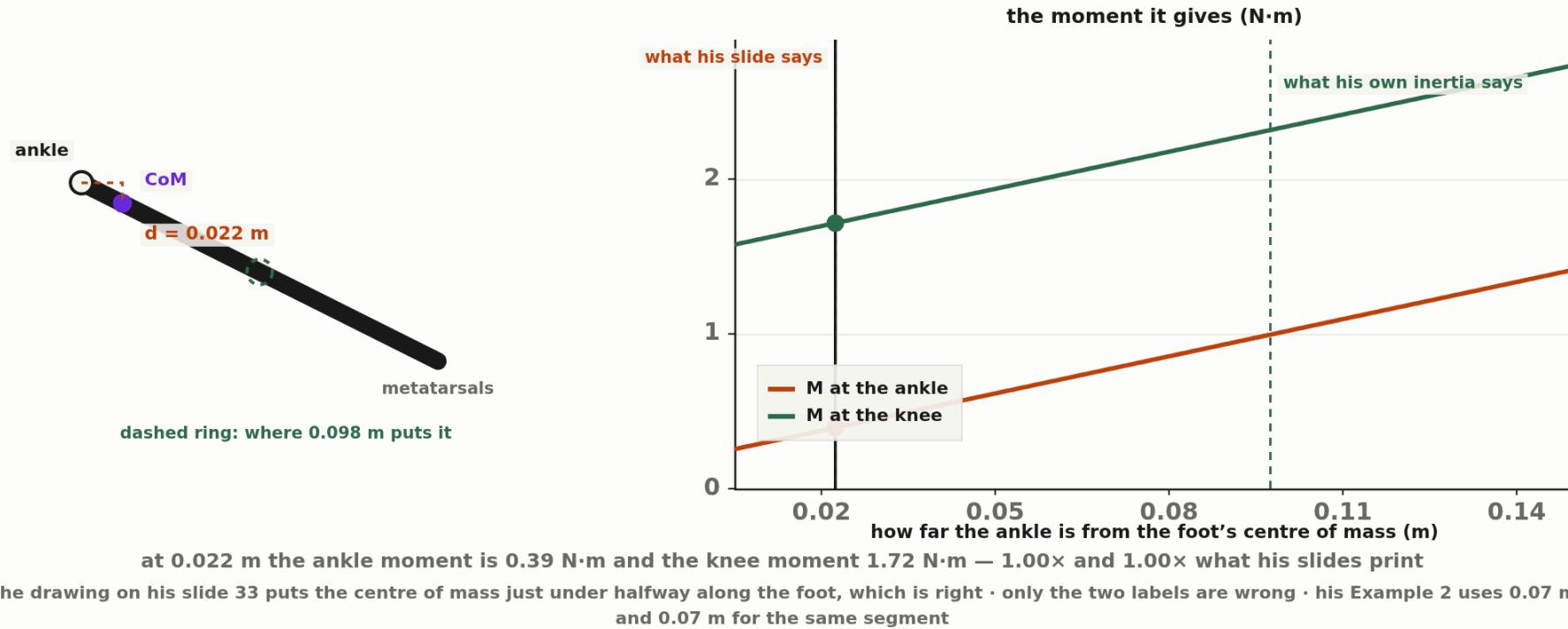
Shank: forces

Shank: moment

The same four steps, twice. In **swing** there is no ground force at all, so the ankle moment is nothing but the segment’s own inertia and the force at its top end: 0.39 N·m. In **stance** the ground pushes with 450 N a hand’s breadth from the centre of mass and the same ankle moment is -76 N·m, two hundred times larger. Nothing about the method changed between them.

# A Worked Example

*The one number that cannot be right*



Ankle to the  
foot's CoM

0.022 m

Everything else on his slides 30 to 41 is right to the digit. This one number is not, and it is not a typo you can shrug at: the ankle moment goes from **0.39** to about **1.0 N·m** and the knee moment from **1.71** to about **2.3**. Three of his own numbers — the foot's moment of inertia, the arms in his Example 2, and the figure he drew — all agree with each other and disagree with the labels. That is worth knowing not because the slide is wrong but because of what it tells you about inverse dynamics: the equations are the easy part, and the anatomy you feed them is where the answer actually comes from.

## On a real stride

The same six lines, run on measured data, and checked against somebody else's answer.

## *The recording*

One stride of overground walking. A woman, 55 kg, 1.61 m, walking at 1.14 m/s with a cadence of 106 steps a minute and a stride length of 1.31 m.

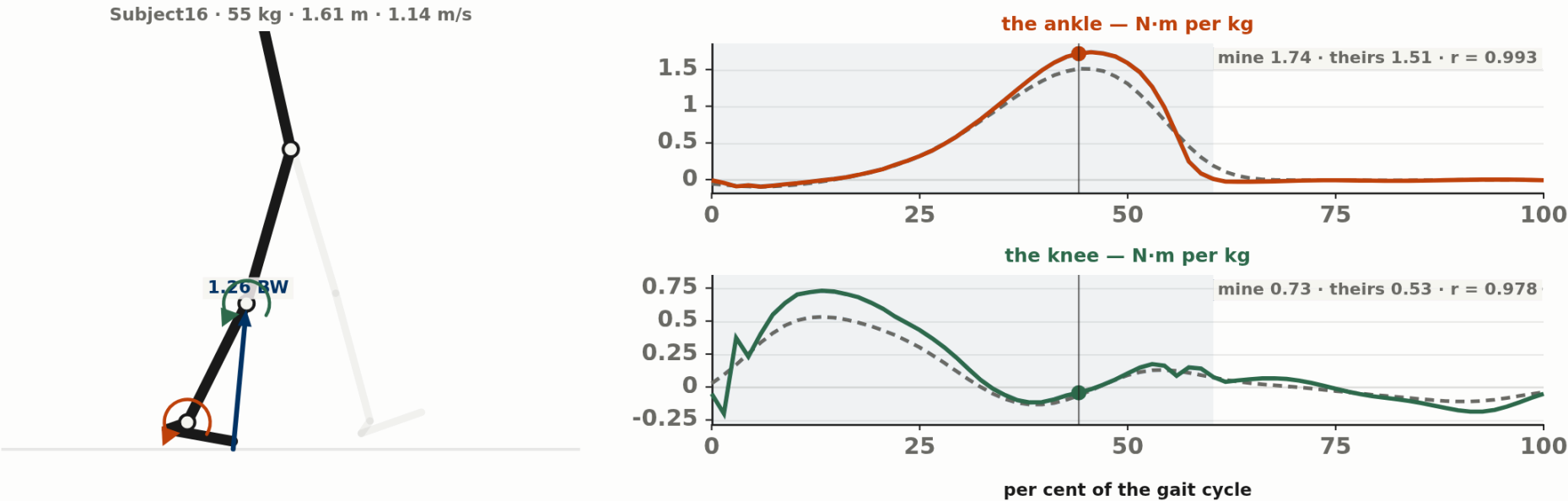
Markers at 60 Hz; the ground reaction force and the centre of pressure at 960 Hz. The plate is under the right foot for the first 60 % of the cycle. Peak vertical force 678 N, which is 1.26 body weights.

Lencioni *et al.* (2019), *Scientific Data* **6**:309, CC BY 4.0 — the same recording lectures 16 and 19 use.

The reason for using this one: the authors **published their own joint moments** for the same stride, computed with their own three-dimensional model. So we can do it from scratch and then mark our own work against somebody who was not trying to agree with us.

Segment masses and inertias from Winter's table, joint centres taken at the skin markers, every derivative taken on a periodically extended record so the seam of the stride does not produce an artefact.

## The same method, run for real



Shaded: the plate is under this foot · dashed: what the authors published for this same stride · solid: the same stride put through the method on his slides · the plate record opens 0.4 BW into loading, so the first two frames are an artefact, not mechanics

Where in the stride

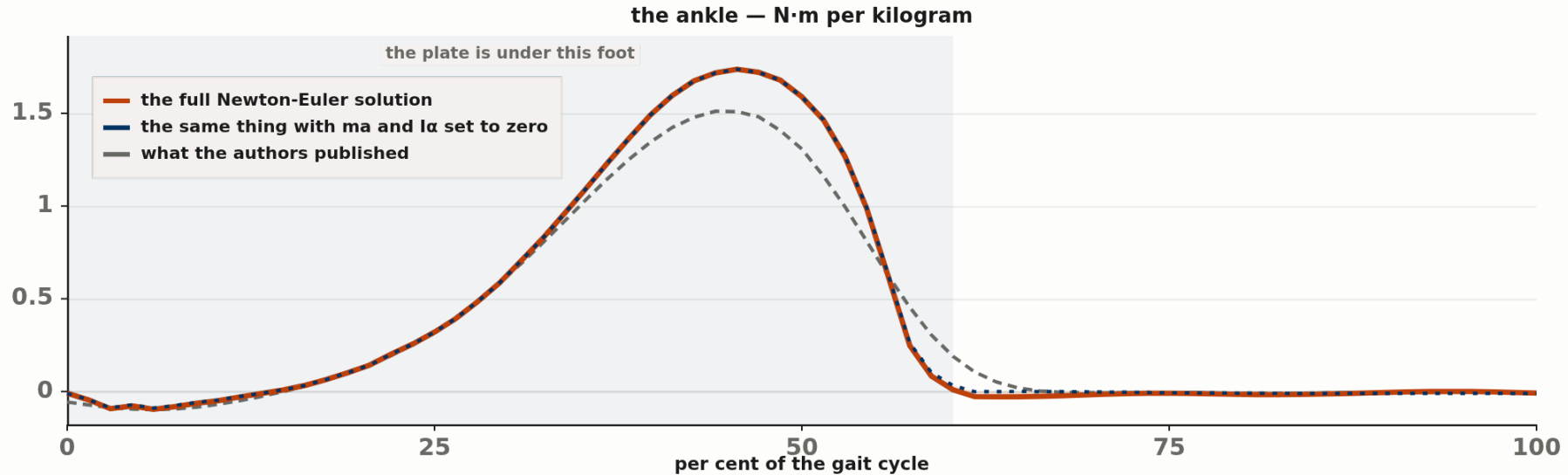
44 %

Both Mine only Theirs only

Walk

One stride, two answers. The shapes match —  $r = 0.99$  at the ankle — and both peaks land on the same instant, while mine run 15 % high at the ankle and 37 % at the knee. No time shift, no centre-of-pressure correction and no single joint-centre nudge closes that gap without making early stance worse: it is the difference between two defensible models of the same body, which is the first three lines of his slide 29.

*So how much of this is dynamics?*



**the dynamics are worth 0.06 % here**

Over stance the two solutions differ by at most 0.023 N·m/kg. At the push-off peak the full solution gives 1.738 and statics gives 1.737 — a difference of 0.06 %. The foot weighs 0.8 kg and is barely accelerating; the ground is pushing with 678 N a hand's breadth from the ankle. Nothing else could matter.

The ankle

The knee

This is the figure lecture 19 asked for. Last week the walking ankle was solved as a **statics** problem and came out 15 % above the published answer, and the obvious suspect was the missing  $ma$  and  $la$ . It was not them. Setting every dynamic term to zero moves the ankle's push-off peak by six hundredths of one per cent. The 15 % is the model — whose anthropometric table, and where you decide the joint is. At the knee in swing it is the other way round entirely, and that is the case worth remembering: **no external force, a heavy segment, and dynamics is the whole of the answer.**

## *What inverse dynamics cannot tell you*

**It gives you:** the net joint moment, the net joint force and the joint power, at every joint, at every instant.

**It cannot give you:**

- which muscles produced the moment, or how hard any one of them pulled;
- the total muscular work, because co-contraction is invisible to it — two muscles pulling hard against each other sum to nothing;
- the bone-on-bone contact force, which is far larger than the net joint force and is not the same quantity.

**And it assumes:** frictionless joints; rigid segments with masses and inertias taken from a table built on cadavers; joint centres estimated from markers on skin that slides over bone; and perfect alignment, in space and in time, between the camera system and the force plate.

Those assumptions are not hypothetical. We just watched them cost fifteen per cent at the ankle and thirty-seven at the knee on a clean recording of ordinary walking.

# **Practice**

Twelve questions, two equations, and one free body diagram every time.

## The Tutor, Dynamic Equilibrium

### Dynamic Equilibrium, question 1

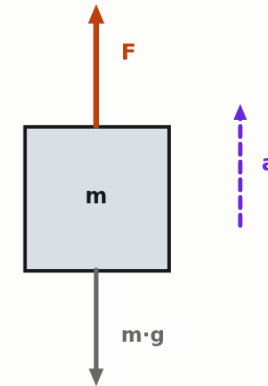
**Landing in running: the whole body centre of gravity accelerates upward at 15 m/s<sup>2</sup>. The man is 75 kg. What vertical force does the ground apply?**

$\Sigma F_y = m \cdot a$ , with the weight on the diagram too

$$F - m \cdot g = m \cdot a$$

$$F = 75(15 + 9.81) = 1860.75 \text{ N}$$

the Tutor accepts 1860.00 to 1862.00 N — and this answer is inside it



Every question in this section is  $\Sigma F = m \cdot a$  or  $\Sigma M = I \cdot \alpha$  with one unknown. Draw the free body first — the Tutor asks for it as its own step in all twelve — and the rest is arithmetic.

1

2

6

3

4

5

7

8

9

10

11

12

The Tutor's Dynamic Equilibrium section, worked. Ten of the twelve come out exactly as it expects. **Question 5** does not, and the fault is the Tutor's: it accepts only 102 m, which you get by rounding the acceleration to 0.51 before squaring the time. **Question 6** has the right accepted answer but a worked solution that substitutes the wrong acceleration. Both are worth reporting rather than working around.

## *What to take away*

- Dynamic analysis is static analysis with  $ma$  and  $I\alpha$  on the right. The diagram, the table and the conventions are unchanged.
- Three equations means three unknowns, which is why you start at the segment touching the ground and work upward.
- Both the force **and** the moment reverse at every joint on the way up.
- Take moments about each segment's own centre of mass, and pair each force with the arm **perpendicular** to it.
- In stance the ground force dominates everything; in swing the segment's own inertia is all there is.
- A joint moment is a **calculation**, not a measurement, and it is only as good as the anatomy you fed it.

## *Where this goes next*

Multiply a joint moment by the joint's angular velocity and you get **joint power** — how fast that joint is putting energy into the body or taking it out. That one multiplication turns this lecture into the standard clinical description of walking.

And the honest limitation remains: inverse dynamics tells you what the joint did, never which muscle did it. Getting from a net moment to individual muscle forces needs either a measurement we cannot make, or a forward model and an assumption about what the nervous system is trying to optimise.

All done.